



LETTERBOXD TASTE ANALYZER

Interactive analysis of personal movie-watching habits, ratings, and prediction.

712	250	3.20	3.5
watched films	diary entries	average rating	median rating

PYTHON • PANDAS • MATPLOTLIB • SCIKIT-LEARN • STREAMLIT • GIT/GITHUB

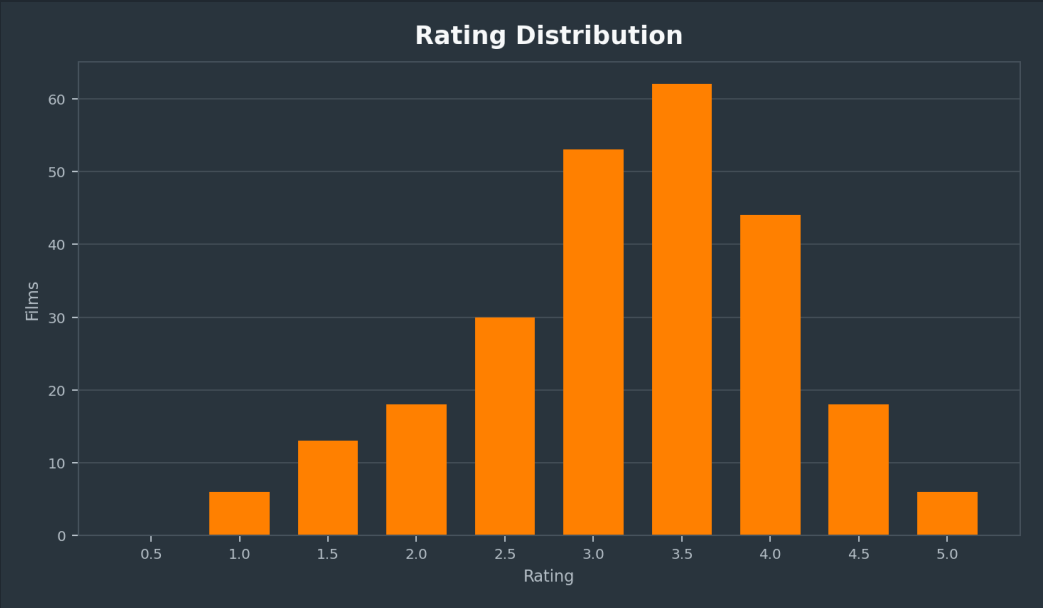


PROJECT OVERVIEW

The app reads a Letterboxd ZIP export, cleans the core CSV files, derives behavioral features, and displays the results in Streamlit.

INPUT	PROCESS	OUTPUT
watched.csv diary.csv ratings.csv	Clean + merge Derive time features Separate rated/unrated	Charts + summary metrics Behavior analysis Ridge baseline

Methodology correction: unrated films remain in viewing totals but are excluded from calculations that require an actual rating.

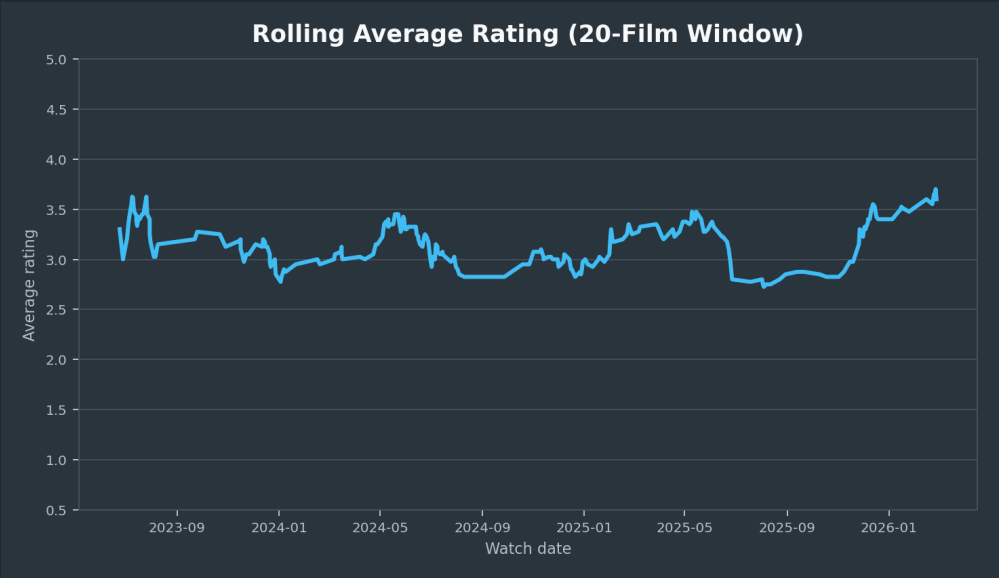


Rating	Count
0.5	0
1.0	6
1.5	13
2.0	18
2.5	30
3.0	53
3.5	62
4.0	44
4.5	18
5.0	6



RATING TREND

A rolling average smooths individual ratings so longer-term changes in how positively I rated movies are easier to see.



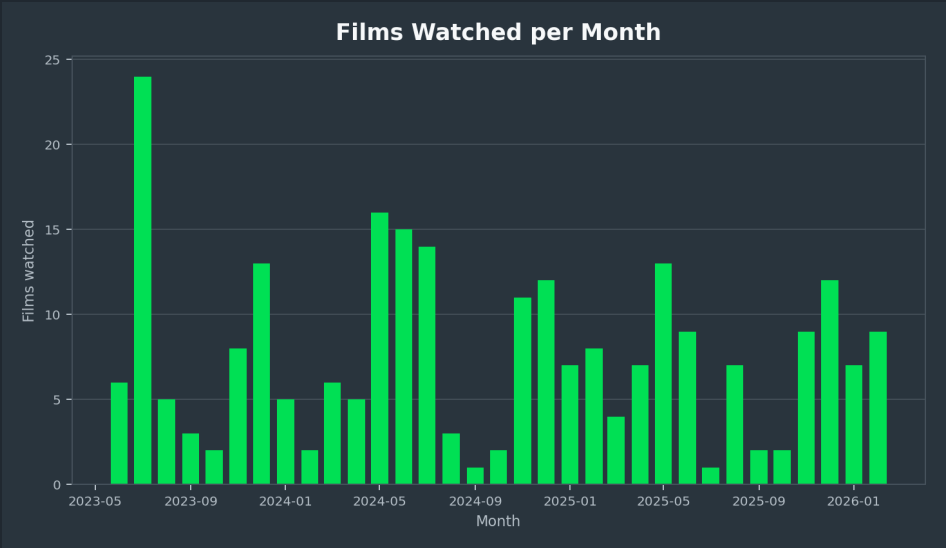
Latest rolling values

Date	Rating	20-film rolling avg
2026-01-19	3.5	3.500
2026-01-20	3.5	3.500
2026-01-25	3.5	3.475
2026-02-15	5.0	3.600
2026-02-22	2.5	3.550
2026-02-23	4.5	3.600
2026-02-24	4.0	3.650
2026-02-25	3.0	3.650
2026-02-25	3.0	3.600
2026-02-25	5.0	3.650
2026-02-26	4.5	3.700
2026-02-27	2.0	3.600



VIEWING VOLUME

Monthly counts show when movie-watching activity increased or slowed down.



Month	Films	Month	Films	Month	Films
2023-06	6	2024-05	16	2025-04	7
2023-07	24	2024-06	15	2025-05	13
2023-08	5	2024-07	14	2025-06	9
2023-09	3	2024-08	3	2025-07	1
2023-10	2	2024-09	1	2025-08	7
2023-11	8	2024-10	2	2025-09	2
2023-12	13	2024-11	11	2025-10	2
2024-01	5	2024-12	12	2025-11	9
2024-02	2	2025-01	7	2025-12	12
2024-03	6	2025-02	8	2026-01	7
2024-04	5	2025-03	4	2026-02	9



TASTE BY RELEASE ERA



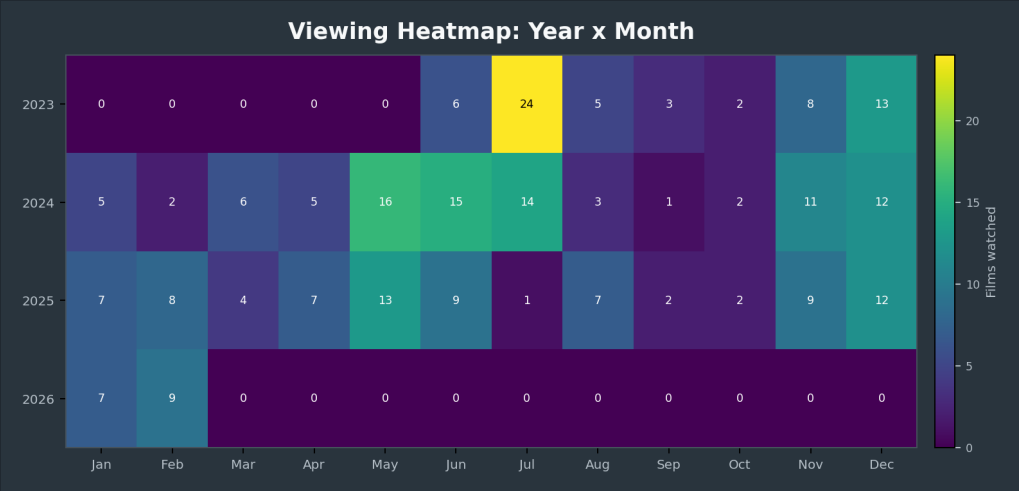
Decade	Films watched	Average rating
1950	1	5.000
1960	5	3.500
1970	6	3.333
1980	18	3.194
1990	19	3.553
2000	36	3.306
2010	66	3.242
2020	99	3.020

Volume and average rating answer different questions. A decade can be watched frequently without necessarily being rated more highly.



WHEN I WATCH

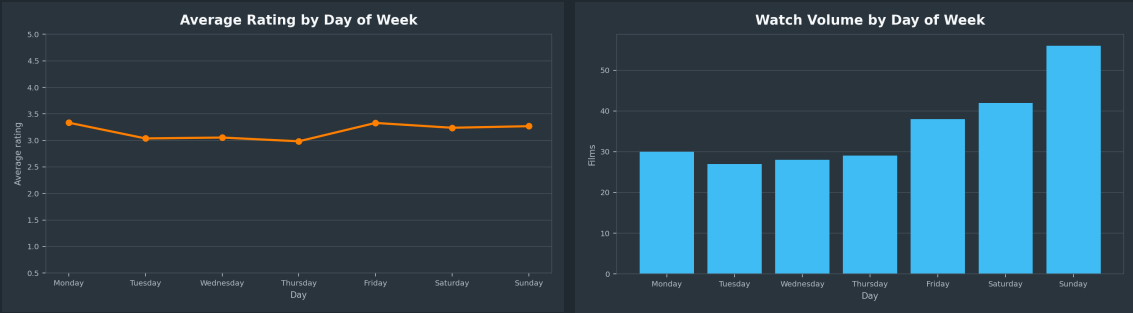
The heatmap shows viewing volume by year and month. The full table below contains the same values, so nothing is hidden or cropped.



Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2023	0	0	0	0	0	6	24	5	3	2	8	13
2024	5	2	6	5	16	15	14	3	1	2	11	12
2025	7	8	4	7	13	9	1	7	2	2	9	12
2026	7	9	0	0	0	0	0	0	0	0	0	0



DAY-OF-WEEK PATTERNS



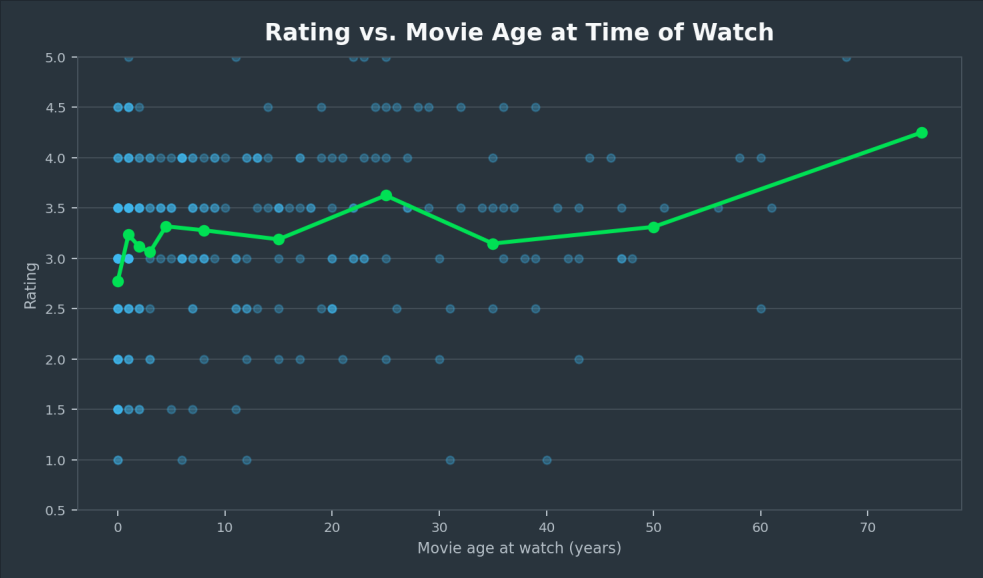
Day	Average rating	Films watched
Monday	3.333	30
Tuesday	3.037	27
Wednesday	3.054	28
Thursday	2.983	29
Friday	3.329	38
Saturday	3.238	42
Sunday	3.268	56

This separates whether a day is associated with different ratings from whether it is simply a more common movie-watching day.



MOVIE AGE AT WATCH

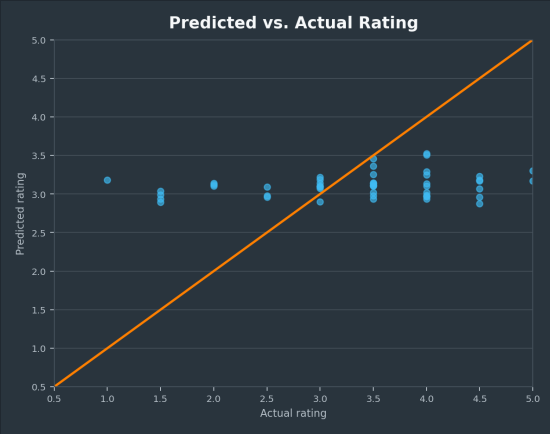
Movie age is calculated as watch year minus release year. The scatter shows individual rated films, while the line summarizes average ratings across age groups.



Age bucket	Films	Average rating
<=0	42	2.774
1	34	3.235
2	13	3.115
3	8	3.062
4-5	11	3.318
6-10	34	3.279
11-20	45	3.189
21-30	28	3.625
31-40	17	3.147
41-60	16	3.312
61+	2	4.250

PREDICTING MY RATING

A Ridge Regression baseline tests how much rating information is contained in watch timing and release-year features alone.



0.791	0.965	-0.010
MAE	RMSE	R ²

Sample test predictions

Month	DOW	Release	Age	Days prev	Actual	Pred.
7	6	1999	25	1	5.0	3.167
2	2	2004	22	0	5.0	3.305
4	4	2024	1	6	4.5	3.172
7	0	2023	0	1	4.5	2.876
7	6	1998	25	1	4.5	3.068
5	2	2011	14	4	4.5	3.228
7	0	2021	2	5	4.5	2.962
7	4	1984	39	2	4.5	3.182
12	5	2018	6	6	4.0	3.132
5	4	2024	1	2	4.0	3.103

Ridge coefficients

Feature	Coefficient
movie_age	0.1065
release_year	0.0994
days_since_prev_watch	0.0179
watch_month	0.0027
watch_dow	0.0020

The near-zero R² shows that these simple features have little predictive power. That result motivates adding genre, director, cast, runtime, and other movie metadata.



DEBUGGING & ITERATION

Area	Problem	Correction
Unrated values	Could be interpreted as zero	Keep missing; exclude only from rating analysis
Rolling average	Could include artificial zeros	Use rated films only
RMSE	Unsupported squared=False argument	<code>sqrt(mean_squared_error(...))</code>
Streamlit	Deprecated sizing syntax	<code>width="stretch"</code>
pandas	Future groupby warning	Set <code>observed=False</code> explicitly

NEXT STEPS

Metadata	Modeling	Product
Genre Director Cast Runtime Language Country	Cross-validation Compare models Feature importance Recommendation methods	Public deployment Movie-level prediction Refined dashboard hierarchy

Project takeaway: the strongest lesson was not achieving a high prediction score. It was learning where the available data was informative, where it could be misleading, and what additional information would be needed for a stronger model.